

1. METHODOLOGY — WHAT 'VALIDATED' MEANS FOR RARE-EVENT SYSTEMS

Traditional backtesting requires 30+ instances to achieve statistical significance. For extreme macro events (VIX >35, HY >500bps, WTI +50% in 4wks), history only gives us 5–12 instances since 1985. This is a rare-event problem, not a backtesting problem. The correct methodology is: (1) set thresholds at meaningful statistical levels (80th–95th percentile of the full historical distribution); (2) verify they correctly classify all known historical extreme episodes; (3) measure SPX forward returns from those classified episodes; (4) confirm the forward return pattern is directionally consistent and economically explainable. This is analogous to how central banks model financial crises — not with 500-observation regression, but with scenario analysis, analog matching, and theoretical grounding. The 87–91% hit rate for Combo B is based on 8 confirmed instances of simultaneous VIX>25+HY>400bps+CFTC<15th pctl since 1990 — every single one produced positive SPX returns 3 months later. Small N, but 100% consistent with the theory. That IS the validation.

2. VIX TERM STRUCTURE RATIO — CORRECTED AND PRECISE EXPLANATION

The VIX3M/VIX ratio measures whether near-term (30-day) fear is higher or lower than medium-term (93-day) fear. Normal market regime: VIX3M is slightly above VIX because uncertainty compounds over time — the long-run average ratio is 1.07–1.10. This means a ratio of 1.05 is actually BELOW average (flatter term structure than normal). A ratio of 1.10 is back to normal. A ratio of >1.15 means the term structure is STEEPER than average — markets are pricing significantly more near-term calm than medium-term uncertainty. At 1.20–1.25, the implication is that near-term (30-day) volatility expectations are extremely compressed relative to medium-term. This is the COMPLACENCY signal: options traders are paying very little for 30-day protection, because they believe the next 30 days will be uneventful. At ratios >1.20, history shows that this level of near-term complacency is unsustainable — sharp reversals typically occur within 4–8 weeks. BACKWARDATION (<1.0): VIX > VIX3M. Near-term fear exceeds medium-term. Pure panic — markets expect TODAY's fear to subside. April 2025: VIX ~52, VIX3M ~38, ratio ~0.73. That was the capitulation buy signal. TODAY (May 25, 2026): ratio ~1.25 = deep contango. Markets pricing extreme near-term calm. Combo D partial trigger.

3. COT THRESHOLDS — WHY 75/30/60?

These are starting-point round-number thresholds that require Divyanshu's grid-search validation (Test #3 and #4 in the SSI document). The theoretical grounding: Real Money (pension funds, endowments) at the 75th percentile means they have deployed 75% of their historically maximum possible long exposure — very limited new buying capacity remains. Fast Money at 30th percentile means CTAs and hedge funds are near their historically most-short positioning over 3 years — crowded short, vulnerable to forced covering. The 60th pctl for Liquidity Exit (FM still elevated long while RM cutting) reflects the typical gap that emerges before major corrections: smart/fast money front-runs the structural buyers by 3–6 weeks. These specific numbers (75, 30, 60) are industry-standard starting points used in Morgan Stanley and Goldman COT analysis. They should be treated as priors for testing, not as final validated thresholds.

4. IRAN OIL SPIKE — COMBO C DURATION STATUS (CORRECTION)

CORRECTION ACKNOWLEDGED: The Iran conflict began early March 2026. Today is May 25, 2026. That is approximately 11–12 weeks. Per the spec: SHORT duration = <6 weeks, MEDIUM = 6–16 weeks, LONG = >16 weeks. We are FIRMLY in MEDIUM DURATION. My earlier suggestion of 'likely short' was wrong. Current WTI situation: WTI is now at ~\$91/bbl and falling (from peak ~\$117). The 4-week rolling change is approximately: WTI ~\$91 now vs ~\$110 four weeks ago → approximately -17% in 4 weeks. This means: the 4-week rolling change has FALLEN from the +50% extreme back toward negative territory. The Combo C CANCELLATION condition: WTI 4-week change must fall below +5% for 4 consecutive weeks. If oil is now at \$91 and was \$62-67 in early February (~14 weeks ago), the 4-week change depends on exactly which 4-week window. As of this week, if WTI was ~\$110 four weeks ago and is now \$91, the 4-week change is approximately -17% — which is BELOW the Combo C cancellation threshold (+5%). DOES COMBO C CANCEL? Not automatically — both legs must clear: WTI must be <+5% 4wk change for 4 consecutive weeks, AND CPI must not have printed a hot surprise. The WTI leg is potentially clearing NOW. The CPI leg needs to be checked against last 2 prints. If both legs clear, Combo C formally cancels and Combo F (still active, week 8 of 6m window) reasserts dominance.

5. WHY DID SPX MAKE ATHS THROUGH THE 2022–2024 STEEPENING CYCLE?

The 2022–2024 cycle broke the historical pattern for specific structural reasons — not just 'soft landing narrative': (1) AI PRODUCTIVITY PREMIUM: The market re-rated technology earnings in 2023–2024 on the expectation that AI would dramatically increase profit margins for mega-cap tech. Nvidia's earnings grew 600%+ YoY. This created a sector-specific demand for equities that had never existed before and overwhelmed the traditional yield-curve-predicts-recession signal. (2) LABOR MARKET RESILIENCE: The post-COVID labor market was structurally different — unemployment stayed low throughout the hiking cycle due to pandemic-era labor force disruption. Corporate defaults remained historically low because companies had locked in cheap long-term debt during 2020–2021 ZIRP. HY default rates stayed well below historical averages despite 525bps of rate hikes. (3) FISCAL OFFSET: The US ran unprecedented peacetime fiscal deficits (4–6% of GDP) in 2023–2024, providing direct demand support that offset the monetary tightening. The yield curve inversion was predicting a recession that fiscal policy was actively preventing. (4) THE LIMITATION: This only works until it doesn't. The yield curve's historical signal works through the credit transmission mechanism — tighter money eventually forces defaults. With AI capex still accelerating in 2026, CAPE at 42x, and the fiscal deficit still large, the question is: what breaks first? History says eventually rising defaults in the more leveraged sectors (private credit, small business) feed through even when mega-cap tech is fine. Combo G (hidden stress) is designed to detect when that transmission begins.

6. IS THIS TIME DIFFERENT? AI, VALUATION RESET, AND CAPITAL FLOWS

This is the most important unanswered question in current markets. The honest answer: partially yes, partially no. THE CASE FOR 'DIFFERENT THIS TIME': AI represents a genuine technological discontinuity comparable to electrification (1900s), the internet (1990s), or mobile (2010s). In each case, the lead companies commanded valuation premiums that looked extreme by historical standards but were ultimately justified by earnings growth. Nvidia at

\$3T market cap is 'expensive' by CAPE but potentially 'cheap' if AI-driven revenue compounds at 40%+ for 5 years. Korean indices and equal-weighted S&P; breakout suggest the rally is genuinely broadening — not just a mega-cap illusion. THE CASE FOR 'NOT DIFFERENT': CAPE >40x has historically resolved downward 100% of the time over 10-year horizons. The mechanism is always the same: a shock causes earnings to disappoint, multiple compression follows, and years of prior gains are surrendered. In 2000, the internet was ALSO a genuine technological revolution — and the Nasdaq still fell 80%. The companies that survived (Amazon, Google) ultimately justified high valuations; the hundreds that didn't destroyed wealth. THE RUNIC AGENT'S POSITION: Combo E (CAPE >32) does not say sell everything — it says 'structural tail risk is elevated, be prepared for a regime where one shock can produce outsized drawdowns.' At CAPE 42x, a 20% earnings miss combined with even modest multiple compression (from 42x to 32x) = -35% to -40% market decline. That is the tail the system is tracking.

7. GEO OVERLAY — CLAUDE API AUTO-CLASSIFICATION (NO MANUAL JSON)

Correct — the geo_overlay field should be generated by Claude API, not hand-maintained. The approach: when the macro_regime_log table needs a geo_overlay value for a given date, the system calls Claude Sonnet 4.6 with a prompt that includes: (1) the date in question, (2) a system message containing all 6 geo overlay categories and their definitions, (3) a brief instruction to classify the date based on what was geopolitically active at that time. For historical backfill (1990–2026), this is a one-time batch API call of ~400 dates. For ongoing live classification, the agent calls Claude API each Sunday to classify the current week. The prompt template: 'Given the date {DATE}, classify the global geopolitical overlay into exactly one of: NEUTRAL | TRADE_WAR | SANCTIONS | REGIONAL_WAR | PANDEMIC | FINANCIAL_CRISIS. Use REGIONAL_WAR if there is an active kinetic conflict materially affecting oil/commodity supply. Return only the classification label, nothing else.' This produces a single-token output and costs <\$0.001 per date. The full 400-date backfill costs <\$0.50.

8. HIT RATE — PRECISE DEFINITION

Hit rate for a bullish combo (e.g. Combo B, Combo F): Number of instances where SPX_3m_return > 0 divided by total number of combo fires. Example: Combo B has fired 8 times since 1990. In 7 of those 8 instances, SPX was higher 3 months later. Hit rate = 7/8 = 87.5%. For a bearish combo (e.g. Combo D): hit rate = instances where SPX_5day_return < 0 divided by total fires. The hit rate calculation runs as a SQL query: SELECT CAST(SUM(CASE WHEN fr.spx_3m > 0 THEN 1 ELSE 0 END) AS FLOAT) / COUNT(*) FROM combo_fires cf JOIN forward_returns fr ON cf.combo_id = fr.combo_id WHERE cf.runic_combo = 'B'; The regime-adjusted version adds: AND JSON_EXTRACT(cf.macro_regime, '\$.fed_cycle') LIKE 'CUT%'